Nikolai Slavov

Contact Information

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Academic Laboratory: slavovlab.net
Parallel Squared Institute: parallelsq.org
SCP Center: center.single-cell.net

Education

Princeton University, Princeton, USA

Ph.D. in Molecular and Quantitative Cell Biology — **06.2010**

M.A. in Molecular Biology — **02.2006**

Mentor: David Botstein

Dissertation: Universality, specificity and regulation of *S. cerevisiae* growth rate response in different carbon sources and nutrient limitations

Massachusetts Institute of Technology (MIT), Cambridge, USA

B.S. in Biology — **06.2004**

Research & Academic Appointments

Parallel Squared Technology Institute (PTI), MA, USA

Founding Director — **2023–present**

Northeastern University, Boston, USA

Professor, Department of Bioengineering — **2024–present**

Faculty Fellow, Barnett Institute — **2017–present**

Affiliated Faculty, Department of Chemistry and Chemical Biology — **2016–present**

Affiliated Faculty, Department of Biology — **2016–present**

Associate Professor, Department of Bioengineering — **2021–2024**

Assistant Professor, Department of Bioengineering — **2015–2021**

Harvard University, Cambridge, USA

Research Fellow, Departments of Systems Biology and Statistics — **2013–2015**

Massachusetts Institute of Technology (MIT), Cambridge, USA

Research Fellow, van Oudenaarden laboratory, Departments of Biology and Physics — **2011–2013**

Awards & Fellowships

- ✓ HUPO Discovery in Proteomic Sciences Award — **2022**
- ✓ College of Engineering Faculty Fellow — **2022**
- ✓ Allen Distinguished Investigator ($ 1, 500, 000) — **2020**
- ✓ Excellence in Mentoring Award — **2018**
- ✓ NIH Director's New Innovator Award ($ 2, 352, 750) — **2016**
- ✓ Broad Institute of Harvard and MIT SPARC Award — **2014**
- ✓ Princeton University Dean's Award — **2010**
- ✓ IRCSET Postgraduate Research Fellowship (72, 000 €) — **2007**
- ✓ Finalist in the Young European Entrepreneur Competition — **2006**

✓ Princeton Graduate Fellowship ($ 45, 000) **2004**

✓ MIT Undergraduate Fellowship ($ 125, 000) **2001**

✓ Eureka Fellowship for Academic Excellence (10, 000 €) **2000**

✓ Bronze Medal in the 31^{st} International Chemistry Olympiad (*IChO*) **1999**

✓ National Diploma for Exceptional Achievements in Chemistry **1999**

Selected Peer Reviewed Publications

[OA] – Open Access article ✉ – Corresponding author Google Scholar Profile

2025 **2025**

✓ **Slavov N**✉
Unlocking the Potential of Single-Cell Omics
J. Proteome Res. 2025, 24, 1481 – 1481 DOI: 0.1021/acs.jproteome.5c00197

✓ Leduc A, Xu Y, Shipkovenska G, Dou Z, **Slavov N**✉
Limiting the impact of protein leakage in single-cell proteomics
Nature Communications (in press) DOI: 10.1038/s41467-025-56736-7

✓ Rengarajan S, Derks J, Bellott D, **Slavov N**, Page DC
Post-transcriptional cross- and auto-regulation buffer expression of the human RNA helicases DDX3X and DDX3Y *Genome Research*, DOI: 10.1101/gr.279707.124

2024 **2024**

✓ Leduc A., Koury L., Cantlon J., Khan S., **Slavov N**✉
Massively parallel sample preparation for multiplexed single-cell proteomics using nPOP
Nature Protocols, DOI: 10.1038/s41596-024-01033-8

✓ SenNet Consortium, **Slavov N**, *et al.*
SenNet recommendations for detecting senescent cells in different tissues
Nat Rev Mol Cell Biol 25(12):1001-1023, DOI: 10.1038/s41580-024-00738-8

✓ Galatidou S, Petelski A, Pujol A, Lattes K,..., **Slavov N**, Barragan M
Single-cell proteomics reveals decreased abundance of proteostasis and meiosis proteins in advanced maternal age oocytes
Molecular Human Reproduction, DOI: 10.1093/molehr/gaae023 **[OA]**

✓ Rogers ZJ, Colombani T, Khan S,..., **Slavov N**, Bencherif SA
Controlling Pericellular Oxygen Tension in Cell Culture Reveals Distinct Breast Cancer Responses to Low Oxygen Tensions
Adv Sci. e2402557, DOI: 10.1002/advs.202402557 **[OA]**

✓ Sipe SS, **Slavov N**✉
Single-Cell Proteomics Accelerates toward Proteoforms
Journal of Proteome Research, DOI: 10.1021/acs.jproteome.4c00290

✓ Khan S, Conover R, Asthagiri AR, **Slavov N**✉
Dynamics of single-cell protein covariation during epithelial-mesenchymal transition
Journal of Proteome Research, DOI: 10.1021/acs.jproteome.4c00277 **[OA]**

2023 **2023**

✓ Wallmann G., Leduc A., **Slavov N**✉
Data-Driven Optimization of DIA Mass Spectrometry by DO-MS
Journal of Proteome Research, DOI: 10.1021/acs.jproteome.3c00177 **[OA]**

✓ **Slavov N**✉
Single-cell proteomics: quantifying post-transcriptional regulation during development with mass-spectrometry
Development, 150 (13): dev201492, DOI: 10.1242/dev.201492

✓ van den Berg PR, Bérenger-Currias NMLP, Budnik B, **Slavov N**✉, Semrau S
Integration of a multi-omics stem cell differentiation dataset using a dynamical model
PLOS Genetics **19** (5) :1-23 DOI: 10.1371/journal.pgen.1010744 **[OA]**

✓ Huffman RG, Leduc A, Wichmann C, di Gioia M, Borriello F, Specht H, Derks J, Khan S,..., **Slavov N**✉
Prioritized mass spectrometry increases the depth, sensitivity and data completeness of single-cell proteomics
Nature Methods, **20**, 714-722 DOI: 10.1038/s41592-023-01830-1 **[OA]**

✓ Gatto L, Aebersold R, Cox J, Demichev V, Derks J, Emmott E, Franks AM, Ivanov AR,..., **Slavov N**✉
Initial recommendations for performing, benchmarking, and reporting single-cell proteomics experiments
Nature Methods **20**, 375–386 DOI: 10.1038/s41592-023-01785-3 **[OA]**

✓ MacCoss MJ, Alfaro J, Faivre, D, Wu C, Wanunu M, **Slavov N**✉
Sampling proteomes by emerging single-molecule and mass-spectrometry methods
Nature Methods **20**, 339–346, DOI: 10.1038/s41592-023-01802-5

✓ **Slavov N**✉
Great Gains in Mass Spectrometry Data Interpretation
Journal of Proteome Research, **22** 3, 659 DOI: 10.1021/acs.jproteome.3c00099

✓ Derks J, **Slavov N**✉
Strategies for increasing the depth and throughput of protein analysis by plexDIA
Journal of Proteome Research **22** 697-705, DOI: 10.1021/acs.jproteome.2c00721 **[OA]**

2022 **2022**

✓ Leduc A, Huffman RG, Cantlon J, Khan S, **Slavov N**✉
Exploring functional protein covariation across single cells using nPOP
Genome Biology, **23**, 261 DOI: 10.1186/s13059-022-02817-5 **[OA]**
↪ Single-Cell Proteomics Bypasses Bottlenecks, Sways Skeptics, highlight by *GEN*

- ✓ Petelski AA, **Slavov N**, Specht H
  Single-Cell Proteomics Preparation for Mass Spectrometry Analysis Using Freeze-Heat Lysis and an Isobaric Carrier
  J. Vis. Exp., (190), DOI: 10.3791/63802

- ✓ Derks J, Leduc A, Wallmann G, Huffman G, Willetts M, Khan S, Specht H,..., **Slavov N**✉
  Increasing the throughput of sensitive proteomics by plexDIA
  Nature Biotechnology, DOI: 10.1038/s41587-022-01389-w
  ↪ Sensitive protein analysis with plexDIA, highlight by *Nature Methods*
  ↪ Uncovering Proteomic Patterns One Cell at a Time, highlight by *GEN*

- ✓ Specht H, **Slavov N**✉
  Beyond Protein Sequence: Protein Isomerization in Alzheimer's Disease
  Journal of Proteome Research 21 (2) :299-300, DOI: 10.1021/acs.jproteome.2c00016

- ✓ SenNet Consortium, **Slavov N**
  NIH SenNet Consortium to map senescent cells throughout the human lifespan to understand physiological health
  Nat aging 1090–1100 DOI: 10.1038/s43587-022-00326-5

- ✓ **Slavov N**✉
  Counting protein molecules for single-cell proteomics
  Cell 185 (2) :232-234, DOI: 10.1016/j.cell.2021.12.013

- ✓ **Slavov N**✉
  Learning from natural variation across the proteomes of single cells
  PLoS Biology 20(1): e3001512, DOI: 10.1371/journal.pbio.3001512 **[OA]**

- ✓ **Slavov N**✉
  Scaling up single-cell proteomics
  Molecular & Cellular Proteomics 21(1), 100179, DOI: 10.1016/j.mcpro.2021.100179 **[OA]**

2021 **2021**

- ✓ Khoury L, **Slavov N**✉
  Comprehensive Identification of Regulatory Protein Networks
  Journal of Proteome Research 20(11): 4913-4914, DOI: 10.1021/acs.jproteome.1c00813

- ✓ He L, *et al.*, **Slavov N**, Marneros A,
  Global characterization of macrophage polarization mechanisms and identification of M2-type polarization inhibitors
  Cell Reports 37(5):109955, DOI: 10.1016/j.celrep.2021.109955

- ✓ Petelski A, Emmott E, Leduc A, Huffman RG, Specht H, Perlman D, **Slavov N**✉
  Multiplexed single-cell proteomics using SCoPE2
  Nature Protocols 16, 5398–5425, DOI: 10.1038/s41596-021-00616-z

- ✓ **Slavov N**✉
  Driving Single Cell Proteomics Forward with Innovation
  Journal of Proteome Research 20, 11, 4915–4918, DOI: 10.1021/acs.jproteome.1c00639

- ✓ **Slavov N**✉
  Measuring Protein Shapes in Living Cells
  Journal of Proteome Research DOI: 10.1021/acs.jproteome.1c00376

✓ **Slavov N**✉
Increasing proteomics throughput
Nature Biotechnology, 39, 809–810, DOI: 10.1038/s41587-021-00881-z

✓ **Slavov N**✉ *et al.*
Voices of biotech research: Single-cell proteomics
Nature Biotechnology, 39, 281–286, DOI: 10.1038/s41587-021-00847-1

✓ Specht H, Emmott E, Petelski A, Huffman RG, Perlman D, Serra M, Kharchenko P, Koller A & **Slavov N**✉
Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity using SCoPE2
Genome Biology, 22, 50 DOI: 10.1186/s13059-021-02267-5 **[OA]**
↪ Towards resolving proteomes in single cells, highlight by *Nature Methods*

2020 **2020**

✓ Specht H, **Slavov N**✉
Optimizing accuracy and depth of protein quantification in experiments using isobaric carriers
Journal of Proteome Research, DOI: 10.1021/acs.jproteome.0c00675

✓ Petelski AA **Slavov N**✉
Analyzing ribosome remodeling in health and disease
Proteomics, 20, (17) :2000039 DOI: 10.1002/pmic.202000039

✓ **Slavov N**✉
Single-cell protein analysis by mass-spectrometry
Current Opinion in Chemical Biology, 60, 1 - 9 DOI: 10.1016/j.cbpa.2020.04.018

✓ **Slavov N**✉
Unpicking the proteome in single cells
Science, 367 (6477) :512 - 513, DOI: 10.1126/science.aaz6695

2019 **2019**

✓ **Slavov N**✉ *et al.*
Voices in methods development: Single-cell proteomics
Nature Methods, 16, 945 - 951, DOI: 10.1038/s41592-019-0585-6

✓ Huffman RG, Chen AT, Specht H & **Slavov N**✉
DO-MS: Data-Driven Optimization of Mass Spectrometry Methods
Journal of Proteome Research, 18, 6, 2493 - 2500, DOI: 10.1021/acs.jproteome.9b00039
↪ Technology feature highlight by *Nature Methods*

✓ Chen AT, Franks A & **Slavov N**✉
DART-ID increases single-cell proteome coverage
PLoS Computational Biology, 5(7):e1007082 DOI: 10.1371/journal.pcbi.1007082 **[OA]**
↪ The single cell proteomics revolution, editorial highlight by *Bioanalysis Zone*

✓ Emmott EP, Jovanovic M & **Slavov N**✉
Approaches for studying ribosome specialization
Trends in Biochemical Sciences, 44(5):478 - 479, DOI: 10.1016/j.tibs.2019.01.008

✓ Emmott EP, Jovanovic M & **Slavov N**✉
Ribosome stoichiometry: from form to function
Trends in Biochemical Sciences, 44(2):95 - 109, DOI: 10.1016/j.tibs.2018.10.009

✓ Malioutov D., Chen T., Jaffe J., Airoldi E., Budnik B. & **Slavov N**✉
Quantifying homologous proteins and proteoforms
Molecular & Cellular Proteomics, 18(1):162 - 168, DOI: 10.1074/mcp.TIR118.000947

2018 **2018**

✓ Specht H. & **Slavov N**✉
Transformative opportunities for single cell proteomics
Journal of Proteome Research, 17 (8), 2565 - 2571 , DOI: 10.1021/acs.jproteome.8b00257
↪ Innovations in Proteomics: The Drive to Single Cells, editorial highlight
↪ Through the Looking Glass of Single Cell Proteomics, highlight by *Technology Networks*

✓ Budnik B., Levy E., Harmange G. & **Slavov N**✉
SCoPE-MS: mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
Genome Biology, 19(1):161, DOI: 10.1186/s13059-018-1547-5 **[OA]**
↪ SCoPE-MS – We can finally do single cell proteomics!!!, highlight by *Proteomics News*
↪ Researchers Apply Mass Spec to Single-Cell Proteomics, highlight by *GenomeWeb*
↪ Single-cell proteomics highlight by *the NIH Director's office*
↪ Technology feature highlight by *Nature Methods*
↪ The single cell proteomics revolution, editorial highlight by *Bioanalysis Zone*

✓ Levy E. & **Slavov N**✉
Single cell protein analysis for systems biology
Essays in Biochemistry, 62(4):595 - 605, DOI: 10.1042/EBC20180014

2017 **2017**

✓ Franks A., Airoldi E.M., **Slavov N**✉
Post-transcriptional regulation across human tissues
PLoS Computational Biology, 13(5): e1005535, DOI: 110.1371/journal.pcbi.1005535 **[OA]**
↪ How Statistics Weakened mRNA's Predictive Power highlight by *The Scientist*

✓ Saleh D, Najjar M, Zelic M, Shah S, Nogusa S, Polykratis A, Paczosa MK, Gough PJ, Bertin J, Whalen M, Fitzgerald KA, **Slavov N**, Pasparakis M, Balachandran S, Kelliher M, Mecsas J, Degterev A.
Kinase Activities of RIPK1 and RIPK3 Can Direct IFN-β Synthesis Induced by Lipopolysaccharide.
The Journal of Immunology, 198(11):4435-4447, DOI: 10.4049/jimmunol.1601717

2016 **2016**

✓ Klionsky, *et al.*, **Slavov N**, *et al.*
Guidelines for the use and interpretation of assays for monitoring autophagy
Autophagy, vol. 12, issue 1, 1–222, 12(1):1-222, DOI: 10.1080/15548627.2015.1100356

✓ Di Luca A, Hamill RM, Mullen AM, **Slavov N**, Elia G.
Comparative Proteomic Profiling of Divergent Phenotypes for Water Holding Capacity across the Post Mortem Ageing Period in Porcine Muscle Exudate
PLoS One, 11(3):e0150605, DOI: 10.1371/journal.pone.0150605 **[OA]**

2015 **2015**

✓ **Slavov N**✉
Making the most of peer review
eLife, 4:e12708, DOI: 10.7554/eLife.12708 **[OA]**

✓ **Slavov N**✉, Semrau S., Airoldi E.M., Budnik B., van Oudenaarden A.
Differential stoichiometry among core ribosomal proteins
Cell Reports, vol. 13, issue 5, 865–873, DOI: 10.1016/j.celrep.2015.09.056 **[OA]**
↪ All Ribosomes Are Created Equal. Really?, highlight by *Trends in Biochemical Sciences*
↪ New direction for tissue engineering and cancer therapeutics, highlight by *Science X*

✓ Alvarez JR, Zhiqiang B, Xu D, Yuan B, Lo KA, Yoon MJ, Lim YC, Knoll M, **Slavov N**, *et al.*
De-Novo Reconstruction of Adipose Tissue Transcriptomes Reveals Long Non-coding RNA Regulators of Brown Adipocyte Development
Cell Metabolism, vol. 21, issue 5, 764–776, DOI: 10.1016/j.cmet.2015.04.003

Started Slavov Lab at Northeastern University, August 2015

2014 **2014**

✓ Malioutov D.✉ , **Slavov N**✉
Convex Total Least Squares
Journal of Machine Learning Research, W&CP, vol. 32, issue 1, 109–117 **[OA]**

✓ **Slavov N**✉, Budnik B., Schwab D., Airoldi E., van Oudenaarden A.✉
Constant Growth Rate Can Be Supported by Decreasing Energy Flux and Increasing Aerobic Glycolysis
Cell Reports vol. 7, issue 3, 705–714 **[OA]**

2013 **2013**

✓ **Slavov N**✉, Carey, J., Linse, S.✉
Calmodulin transduces Ca^{+2} oscillations into differential regulation of its target proteins
ACS Chemical Neuroscience, vol. 4, 601–612 **[OA]**

✓ **Slavov N**✉, Botstein D.✉
Decoupling Nutrient Signaling from Growth Rate Causes Aerobic Glycolysis and Deregulation of Cell Size and Gene Expression
Molecular Biology of the Cell, vol. 24, issue 2, 157–168 **[OA]**

2012 **2012**

✓ **Slavov N**✉, van Oudenaarden A.✉
How to Regulate a Gene: To Repress or to Activate?
Molecular Cell, vol. 46, issue 5, 551–552 **[OA]**

✓ **Slavov N**✉, Airoldi E.M., van Oudenaarden A., Botstein D.✉
A Conserved Cell Growth Cycle Can Account for the Environmental Stress Responses of Divergent Eukaryotes
Molecular Biology of the Cell, vol. 23, no. 10, 1986–1997 **[OA]**

2011 2011

✓ **Slavov N**✉, Macinskas J., Caudy A., Botstein D.✉
Metabolic Cycling without Cell Division Cycling in Respiring Yeast
PNAS, vol. 108, no. 47, 19090–19095 **[OA]**

✓ **Slavov N**✉, Botstein D.✉
Coupling among Growth Rate Response, Metabolic Cycle and Cell Division Cycle in Yeast
Molecular Biology of the Cell, vol. 22, no. 12, 1997–2009 **[OA]**

2010 2010

✓ **Slavov N**✉
Inference of Sparse networks with Unobserved Variables. Application to Gene Regulatory Networks
Journal of Machine Learning Research, W&CP, vol. 9, 757–764 **[OA]**

✓ **Slavov N**✉, Botstein, D.
Universality, Specificity and Regulation of *S. cerevisiae* Growth Rate Response in Different Carbon Sources and Nutrient Limitations
Dissertation, Princeton University **[OA]**

✓ Silverman SJ., **Slavov N**, Petti A., Parsons L., Briehof R., Thiberge S., Zenklusen D., Gandhi SJ., Larson D., Singer R., Botstein D.✉
Metabolic cycling in single yeast cells from unsynchronized steady-state populations limited on glucose or phosphate
PNAS, vol. 107, no. 15, 6946–6951 **[OA]**

2009 and before 2009 and before

✓ **Slavov N**✉, Dawson KA. (2009)
Correlation Signature of the Macroscopic States of the Gene Regulatory Network in Cancer
PNAS, vol. 106, no. 11, 4079-4084 (“In This Issue” article) **[OA]**

✓ Tagkopoulos I.✉ , **Slavov N**✉, Kung S.✉ (2005)
Multi-Class Biclustering and Classification Based on Modeling of Gene Regulatory Networks
5^{th} IEEE Symposium on Bioinformatics and Bioengineering (BIBE’05).

PATENTS

- **Slavov N**✉, *et al.*, Miniaturized Proteomic Sample Preparation. U.S. Patent Application No. 18/556,655.
- **Slavov N**✉, *et al.*, Methods of multiplexed data-independent acquisition for proteomics. U.S. Patent Application No. 17/806,468.

- **Slavov N**✉, *et al.*, Mass spectrometry technique for single cell proteomics, US Patent No. 11,719,703 issued on 8/8/2023.

PUBLISHED PREPRINTS
GOOGLE SCHOLAR

- ▷ Leduc A, **Slavov N**✉ (2025)
  Protein degradation and growth dependent dilution substantially shape mammalian proteomes
  bioRxiv DOI: 10.1101/2025.02.10.637566

- ▷ Khan S, Elcheikhali M, Leduc A, Huffman R, Derks J, Franks A, **Slavov N**✉ (2024)
  Inferring post-transcriptional regulation within and across cell types in human testis
  bioRxiv DOI: 10.1101/2024.10.08.617313

- ▷ Iwamoto-Stohl LK, Petelski AA, ... **Slavov N**, Zernicka-Goetz (2024)
  Proteome asymmetry in mouse and human embryos before fate specification
  bioRxiv DOI: 10.1101/2024.08.26.609777

- ▷ Tsour S, Machne R, Leduc A, Widmer S, **Slavov N**✉ (2024)
  Alternate RNA decoding results in stable and abundant proteins in mammals
  bioRxiv DOI: 10.1101/2024.08.26.609665

- ▷ Leduc A, Xu Y, Shipkovenska G, Dou Z, **Slavov N**✉ (2024)
  Limiting the impact of protein leakage in single-cell proteomics
  bioRxiv DOI: 10.1101/2024.07.26.605378

- ▷ Derks J, Jonson T, Leduc A, Khan S, Khoury L, Rafiee M, **Slavov N**✉ (2024) Single-nucleus proteomics identifies regulators of protein transport
  bioRxiv DOI: 10.1101/2024.06.17.599449

- ▷ Galatidou S, Petelski A, Pujol A, ..., **Slavov N**✉, Barragan M (2024)
  Single-cell proteomics reveals decreased abundance of proteostasis and meiosis proteins in advanced maternal age oocytes
  bioRxiv DOI: 10.1101/2024.05.23.595547

- ▷ Khan S, Conover R, Asthagiri AR, **Slavov N**✉ (2023)
  Dynamics of single-cell protein covariation during epithelial-mesenchymal transition
  bioRxiv, DOI: 10.1101/2023.12.21.572913 **[OA]**

- ▷ Leduc A., Koury L., Cantlon J., **Slavov N**✉ (2023)
  Massively parallel sample preparation for multiplexed single-cell proteomics using nPOP
  bioRxiv, DOI: 10.1101/2023.11.27.568927 **[OA]**

- Leduc A, Harens H, **Slavov N**✉ (2023)
  Modeling and interpretation of single-cell proteogenomic data
  arXiv DOI: 10.48550/arXiv.2308.07465

- Wallmann G., Leduc A., **Slavov N**✉ (2023)
  Data-Driven Optimization of DIA Mass-Spectrometry by DO-MS
  bioRxiv, DOI: 10.1101/2023.02.02.526809 **[OA]**

- Derks J, **Slavov N**✉ (2022)
  Strategies for increasing the depth and throughput of protein analysis by plexDIA
  bioRxiv, DOI: 10.1101/2022.11.05.515287 **[OA]**

- Gatto L, ..., **Slavov N**✉ (2022)
  Initial recommendations for performing, benchmarking, and reporting single-cell proteomics experiments
  arXiv, DOI: 10.48550/arXiv.2207.10815 **[OA]**

- MacCoss MJ, Alfaro J, Wanunu M, Faivre DA, **Slavov N**✉ (2022)
  Sampling proteomes by emerging single-molecule and mass-spectrometry methods
  arXiv, DOI: 10.48550/arXiv.2208.00530 **[OA]**

- Huffman RG, Leduc A, ..., **Slavov N**✉ (2022)
  Prioritized single-cell proteomics reveals molecular and functional polarization across primary macrophages
  bioRxiv, DOI: 10.1101/2022.03.16.484655 **[OA]**

- Derks J, Leduc A, Huffman RG, Specht H, Ralser M, Demichev V, **Slavov N**✉ (2021)
  Increasing the throughput of sensitive proteomics by plexDIA
  bioRxiv, DOI: 10.1101/2021.11.03.467007 **[OA]**

- Leduc A, Huffman RG, **Slavov N**✉ (2021)
  Droplet sample preparation for single-cell proteomics applied to the cell cycle
  bioRxiv, DOI: 10.1101/2021.04.24.441211 **[OA]**

- Petelski A, Emmott E, Leduc A, Huffman RG, Specht H, Perlman D, **Slavov N**✉ (2021)
  Multiplexed single-cell proteomics using SCoPE2
  bioRxiv, DOI: 10.1101/2021.03.12.435034 **[OA]**

- Specht H, **Slavov N**✉ (2020)
  Optimizing accuracy and depth of protein quantification in experiments using isobaric carriers
  bioRxiv, DOI: 10.1101/2020.08.24.264994 **[OA]**

- Petelski AA, **Slavov N**✉ (2020)
  Analyzing ribosome remodeling in health and disease
  arXiv, arXiv:2007.05839 **[OA]**

▷ **Slavov N**✉ (2020)
Single-cell protein analysis by mass-spectrometry
arXiv, arXiv:2004.02069 **[OA]**

▷ Specht H, Emmott E, Petelski A, Huffman RG, Perlman D, *et al.* & **Slavov N**✉ (2019)
Single-cell mass-spectrometry quantifies the emergence of macrophage heterogeneity *bioRxiv*, DOI: 10.1101/665307 **[OA]**
↪ Over 1,000 SINGLE CELL PROTEOMES!, highlight by *Proteomics News*
↪ Technology feature highlight by *Nature Methods*

▷ Specht H, Emmott E, Koller A, **Slavov N**✉ (2019)
High-throughput single-cell proteomics quantifies the emergence of macrophage heterogeneity
bioRxiv, DOI: 10.1101/665307 **[OA]**

▷ Huffman G, Specht H, Chen AT, **Slavov N**✉ (2019)
DO-MS: Data-Driven Optimization of Mass Spectrometry Methods
bioRxiv, DOI: 10.1101/512152 **[OA]**

▷ Specht H, Harmange G, Perlman DH, Emmott E, Niziolek Z, Budnik B, **Slavov N**✉ (2018)
Automated sample preparation for high-throughput single-cell proteomics
bioRxiv, DOI: 10.1101/399774 **[OA]**

▷ Chen A, Franks A, **Slavov N**✉ (2018)
DART-ID increases single-cell proteome coverage
bioRxiv, DOI: 10.1101/399121 **[OA]**

▷ Levy E. & **Slavov N**✉ (2018)
Single cell protein analysis for systems biology
PeerJ Preprint, DOI: 10.7287/peerj.preprints.26965v1 **[OA]**

▷ Emmott EP, Jovanovic M & **Slavov N**✉ (2018)
Ribosome stoichiometry: from form to function
PeerJ Preprint, DOI: 10.7287/peerj.preprints.26991v1 **[OA]**

▷ Specht H. & **Slavov N**✉ (2018)
Routinely quantifying single cell proteomes: A new age in quantitative biology and medicine
PeerJ Preprint, DOI: 10.7287/peerj.preprints.26821v1 **[OA]**

▷ Malioutov D., Chen T., Jaffe J., Airoldi E., Carr S., Budnik B., **Slavov N**✉ (2017)
Quantifying homologous proteins and proteoforms
bioRxiv, DOI: 10.1101/168765 **[OA]**

▷ Berg P., Budnik B., **Slavov N**✉, Semrau S. (2017)
Dynamic post-transcriptional regulation during embryonic stem cell differentiation
bioRxiv, DOI: 10.1101/123497 **[OA]**

▹ Budnik B., Levy E., **Slavov N**✉ (2017)
Mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
bioRxiv, DOI: 10.1101/102681 **[OA]**
↪ SCoPE-MS – We can finally do single cell proteomics!!!, highlight by *Proteomics News*
↪ Researchers Apply Mass Spec to Single-Cell Proteomics, highlight by *GenomeWeb*
↪ Interview, highlight by *Front Line Genomics*

▹ **Slavov N**✉ (2015)
From differential transcription of ribosomal proteins to differential structure of ribosomes
PeerJ Preprint, DOI: 10.7287/peerj.preprints.1504v1 **[OA]**

▹ Franks A., Airoldi E.M., **Slavov N**✉ (2015)
Post-transcriptional regulation across human tissues
bioRxiv, DOI: 10.1101/020206 **[OA]**

▹ **Slavov N**✉, Botstein, D., Caudy A. (2014)
Extensive regulation of metabolism and growth during the cell division cycle
bioRxiv, DOI: 10.1101/005629 **[OA]**

▹ **Slavov N**✉, Semrau S., Airoldi E.M., Budnik B., van Oudenaarden A. (2014)
Differential stoichiometry among core ribosomal proteins
bioRxiv, DOI: 10.1101/005553 **[OA]**

Opinions & Popular Science Publications

▹ **Slavov N**✉ (2018) We Must Demand Evidence of Peer Review
The Scientist, Link

▹ **Slavov N**✉ (2017) What was rate limiting was the idea, not the technology
Front Line Genomics, Link

▹ **Slavov N**✉ (2015) Making the most of peer review
eLife, Link

▹ **Slavov N**✉ (2014) Accomplishments Over Accolades!
The Scientist, Link

▹ **Slavov N**✉ (2014) Discoveries lie hidden behind the façade of popular assumptions
Cell Reporter, Link

▹ **Slavov N**✉ (2013) The Best Projects Are Least Obvious
The Scientist, Link

▹ **Slavov N**✉ (2013) A Wonderful Christmas with Mysterious Oscillations
PubChase, Link

▷ **Slavov N**✉ (2012) The Mission of MIT
The MIT Tech, Link